

Yash Rajesh Hiray

Data & Applied AI Engineer | Python | Databricks | PySpark | RAG, LangChain | Agentic AI | FastAPI | AWS

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PROFESSIONAL SUMMARY

Data & Applied AI Engineer designing production-grade data platforms and AI applications - spanning ELT pipeline engineering, data modeling, LLMs, Retrieval-Augmented Generation (RAG), and Agentic AI. Proven expertise in cloud-native systems (Python, PySpark, Databricks, FastAPI, PostgreSQL, AWS) - delivered AI solutions processing 100K+ enterprise documents and modernized data platforms supporting 25M+ records across global consulting engagements.

TECHNICAL SKILLS

Data & Infrastructure: Python · SQL · Databricks · Apache Spark (PySpark) · ETL/ELT Pipelines · Medallion Architecture · PostgreSQL · Data Modeling & Quality · AWS (S3, Lambda, Bedrock) · Docker · CI/CD · FastAPI
AI/ML & Agentic Systems: LLMs · MCP · RAG Pipelines · LangChain · LLM Fine-tuning (LoRA) · Vector Search & Embeddings · Multi-agent Orchestration (CrewAI) · Hugging Face · Claude API · Prompt Engineering

WORK EXPERIENCE

McKinsey & Company | Data Engineer I

Chennai, India | 2024 – Present

- Enhanced an enterprise Document Intelligence Platform (React, FastAPI) - an asset analyzing **100K+** documents annually - personally supporting **3** consulting engagements; built multi-modal extraction pipelines using **LayoutLM**, **PaddleOCR**, and **Tesseract**, cutting manual reconciliation effort **5x** and turnaround from days to hours.
- Modernized a legacy benchmarking platform by consolidating **72** fragmented tables into **14** governed **Delta Lake** entities using **Medallion Architecture**, reducing data model complexity by **80%+** across **25M+** records.
- Engineered **12+** production **ELT pipelines** (**PySpark**, **SQL**, **Databricks Jobs**, **GitHub Actions**) with automated data-quality validations, recovering **9,000+** records and delivering the platform's first **Power BI** self-service layer for **20+** stakeholders.
- Led migration of **25M+** records from **Databricks Delta Lake** to **PostgreSQL**, redesigning **14+** entity schemas and building automated validation and reconciliation frameworks (**Python**, **FastAPI**, **Alembic**) to enable a scalable, cloud-native backend for next-generation applications.

McKinsey & Company | Solution Delivery Intern

Chennai, India | Jan 2023 – Jul 2023

- Developed a conversational legal document intelligence chatbot using RAG, LangChain, and vector databases, enabling lawyers to query contract clauses, obligations, and risks in natural language - eliminating manual review.
- Fine-tuned **Flan-T5-XL** with **LoRA** on the **CUAD** legal dataset, achieving **4x** uplift over baseline (**BLEU 0.066** → **0.298**, **ROUGE-1 F1 0.152** → **0.630**) on contract question answering.
- Architected a unified LLM pipeline (PDF extraction → chunking → embeddings → vector search → retrieval) replacing multiple task-specific models, extended with conversational legal document drafting in a single Streamlit interface.

OTHER PROJECTS

- Insurance Buddy - Agentic AI Insurance Recommendation Platform (CrewAI, GPT-4o)** - Built at an internal innovation hackathon; a 4-agent architecture orchestrating autonomous research, critique, and report generation with real-time personalized policy recommendations.
- Intelligent Document Classification Engine** - Engineered an end-to-end NLP classification pipeline for medical and news documents (**OHSUMED**) combining **TF-IDF** with **sciSpaCy NER**; benchmarked **5** classifiers (SVM, XGBoost, Random Forest) with **10-fold cross-validation** to replace manual labelling workflows.

EDUCATION

M.Tech, Data Analytics | NIT Tiruchirappalli (Academic Excellence Award)

CGPA: 9.15/10 | 2021 – 2023

B.Tech, Computer Science | Department of Technology, Shivaji University

CGPA: 8.9/10 | 2016 – 2020

CERTIFICATIONS

AWS Certified Cloud Practitioner -Amazon Web Services | Neural Networks & Deep Learning -DeepLearning.AI

LEADERSHIP & COMMUNITY

- Enabled AI adoption across social impact initiatives by training **110+** **Generation Aid India** associated college students on ATS-ready resumes and interview preparation, and **30+** **visually challenged faculty** at **Help the Blind Foundation** on Gemini Live, Google Workspace AI, and prompt engineering.
- Led a **45+** **member team** executing campus animal welfare programs - sterilization, adoption, and feeding drives with veterinary care - in partnership with Blue Cross Trichy (**Prakruti Nature Club**, **NIT Trichy**).